

Five MariaDB advantages and features

Many features and benefits distinguish MariaDB from MySQL and other open source RDBMS. Let's look at the five aspects that set MariaDB apart from the others.

The InnoDB database: InnoDB is a general-purpose storage engine notable for its ability to balance high dependability and high performance. It is MySQL's default storage engine and a popular choice for usage with MariaDB. It supports foreign keys and provides conventional ACID-compliant transaction capabilities. The acronym ACID stands for atomicity, consistency, isolation, and durability. This ensures that each transaction is handled as a separate entity. If that transaction fails to complete, the entire transaction fails, leaving the database unaltered. InnoDB employs a dual licence, one for use with GNU GPL software and another with proprietary software.

XtraDB (ExtraDB): XtraDB, once a more popular option than InnoDB, was created as a drop-in storage engine for MariaDB. This was in versions before 10.1. InnoDB has been the default storage engine for MariaDB since version 10.2.

RocksDB/MyRocks: MyRocks is open source software created and still maintained by Facebook's database engineering team. It is an alpha-stage storage engine that is optimised for low latency and rapid storage. Its primary purpose is to maximise storage savings. By focusing on efficiency, it helps to save money on things like SSD wear and tear, actual storage required, and greater IO capacity for query management. It is officially supported on CentOS 6.8 and 7.2.X, with best-effort support for Ubuntu 14, 15, and 16 LTS, and subsequent versions.

Cluster of Galera: Galera cluster is a 'real multi-master cluster' that relies on synchronous replication. Its primary goal is to ensure high uptime, prevent data loss, and be scalable for future growth. Synchronous replication ensures that the slave does not lag and that no data is lost if the node fails. It can read and write to any node at any time, and multi-threads the slave for improved speed. It has a 'Hot Standby' feature, which means there is no downtime during a failover because there is no failover.

Sequence storage engine: Sequence engines let you generate ascending or descending numerical sequences with a particular starting, ending, and increment value. As needed, this generates virtual ephemeral tables. There is no method to build a sequence table intentionally, and these engines do not write to a disc or develop from files; they are read-only and transactional.

What exactly is MySQL?

MySQL, created in the mid-1990s, was one of the first open source databases accessible on the market. There are numerous MySQL alternatives available nowadays. However, the differences between the variants are minor because they employ the same syntax and have the same core functionality. MySQL is an RDBMS that aids in the organisation of data in a database. It is utilised on top of a Linux distribution with the combination of PHP and Apache Web Server. To query the database, MySQL employs the SQL programming language.

This database management system aids in adequate storage and management of data. In general, databases store data in a structured manner. It is written in C and C++ and has been tested with several compilers to ensure no errors or inconsistencies.

Important MySQL features

Open source: MySQL is open source software, which implies that anybody can download, use, and modify it. It is free to use and simple to comprehend. MySQL's source code can be examined and modified to meet specific needs. It makes use of the GPL, or GNU General Public License, which outlines the rules and regulations for what can and cannot be done with the application.

Scalable: The ability of systems to work quickly with small amounts of data, vast amounts of data, clusters of machines, and so on is called scalability. MySQL server has been designed to handle massive databases.

Types of data: It supports a wide range of data types, including unsigned and signed integers, float (FLOAT), double (DOUBLE), character (CHAR), variable character (VARCHAR), text, blob, date, time, DateTime, timestamp, year, and so on.

Sets of characters: MySQL supports a variety of character sets, including latin1 (cp1252 character encoding), German, Ujis, and other Unicode character sets.

Secure: It provides a safe interface since it includes a customisable password mechanism to check it, based on the host, before accessing the database. While connecting to the server, the password is encrypted.

Large database support: It supports big databases including 40 to 50 million records, 150,000 to 200,000 tables, and up to 5,000,000,000 rows.

Utility and client programs: MySQL server also includes a slew of client and utility apps. This covers command-line utilities like *mysqladmin* as well as graphical programs like 'MySQL Workbench'. MySQL client programs are written in several different languages. Client libraries (code contained in modules) can be written in C or C++, and are available to clients with C bindings.